

Measurement of H₂O adsorption under Mars-like conditions: Effects of adsorbent heterogeneity

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Abstract. New measurements of the adsorption of H₂O onto terrestrial materials, under Mars-like conditions, disagree with predictions made on the basis of the most frequently used adsorption isotherms. We report here on additional measurements, which confirm that previous estimates of H₂O adsorptive coverage of Martian surface minerals were too high. This discrepancy is a result of extrapolating an empirical expression for adsorption that can be shown to contain unrealistic assumptions. New isotherms, developed to describe adsorption on heterogeneous surfaces, are well suited to describing adsorption at Mars-like conditions. The new isotherms predict that the adsorptive capacity of the regolith is lower than previously thought, if normalized for available surface area. The adsorptive behavior of the regolith materials, to first order, can be predicted on the basis of their specific surface area, without regard to composition. For a given specific surface area of a few tens of square meters per gram, we show that vapor diffusion is more rapid, and that predawn, near-surface ground ice must occur more commonly than previously thought. We find that the presence of a few percent of smectites in the upper few centimeters of the regolith can dominate the adsorption cycle.

1. Introduction

We have made new adsorption measurements with the objective of understanding the role of adsorption in the Martian climate. Our goals in this paper are to (1) describe new measurements of adsorption of H₂O onto Mars analog materials at Mars-like temperatures, (2) compare the measured adsorptive behavior to previously predicted adsorptive behavior, quantitatively and in terms of adsorption theory, and (3) revise the application of H₂O adsorption to Mars and examine the results of the revised theory in terms of boundary layer physics, adsorbate inventory, and diffusion length and timescales.

The physical adsorption of H₂O onto the surface of the solid particulates that constitute the Martian regolith has long been recognized as a potential reservoir of exchangeable volatiles. Water adsorbed onto the regolith is free to exchange within the atmosphere-cap-regolith system. Numerical models of diurnal and seasonal H₂O behavior in the Martian atmosphere are incapable of reproducing the observed behavior of atmospheric H₂O without incorporating a regolith reservoir. The adsorptive capacity of the regolith material determines the timescale over which the regolith can buffer the atmosphere.

Adsorption also acts to slow the diffusion of H₂O through the regolith. In effect the diffusion coefficient must be scaled by the slope of the adsorption isotherm in order to account for the fact that a given molecule spends considerably more time adsorbed onto the walls of the pores than in actual diffusion. The thermal diffusivity through the Martian regolith is greater than the vapor diffusivity, and the thermal wave regularly outruns the H₂O diffusion wave because of the adsorptive slowing of diffusion. The specifics of the balance, however, are contingent on the degree of adsorptive slowing because in the absence on adsorption, vapor diffusion outstrips thermal diffusion.

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Finally, H₂O adsorption in the Martian regolith retards formation of ice by segregating H₂O molecules into another phase. In climate models run without adsorption, H₂O sublimated from the north polar cap quickly condenses across much of the Martian surface, forming extensive ice deposits [Houben *et al.*, this issue]. Likewise, diurnal models predict nighttime surface and subsurface ice without adsorption [Flasar and Goody, 1976].

2. Background

2.1. Previous Studies

The model of adsorption that has been used in the literature for years is based on a set of measurements made by Fanale and Cannon [1971, 1974]. They measured the adsorption of H₂O on basalt. The conditions of their measurements were $T = 250\text{--}300\text{ K}$ and $P = 2\text{--}2000\text{ Pa}$; they found that of the order of $10^{-3}\text{ kg of H}_2\text{O kg}^{-1}$ were adsorbed. These data had to be numerically modeled to apply to Mars, since the conditions under which the measurements were made were quite different from Mars ($T = 150\text{--}300\text{ K}$, $P = 10^{-3}\text{ to }1.0\text{ Pa}$).

Several empirical representations of this data were later developed and used extensively in the literature. The technique involved a least squares fit of an empirical isotherm

$$\rho_a = \rho_s A_s \frac{\gamma P^{0.51}}{\exp(\delta/T)} \quad (1)$$

which represented the available data well. Here, γ is $2.0847 \times 10^{-12}\text{ (kg m}^{-2}\text{ Pa}^{-1})$, $\delta = -2679.8\text{ K}$, ρ_s is the density of the regolith (kg m^{-3}), and A_s is the specific surface area of the regolith ($\text{m}^2\text{ kg}^{-1}$). Equation (1) yields the adsorptive load in kilograms of H₂O m^{-3} regolith. This isotherm was used to estimate the total abundance of H₂O adsorbed in the regolith, the diffusion scale of H₂O over a variety of depths, and the behavior of H₂O in the diurnal boundary layer and in long-

Table 1. Comparison of Conditions

	Mars	Fanale and Cannon [1974]	Zent and Quinn [1995]
Pressure	10 ⁻³ –10 ⁻¹ Pa	2–2000 Pa	10 ⁻² –600 Pa
Temperature	150–300 K	250–290 K	210–290 K

term models of climate evolution [Flasar and Goody, 1976; Toon *et al.*, 1980; Zent *et al.*, 1993; Mellon and Jakosky, 1995].

The use of this isotherm at Mars conditions led to a number of predictions. At average Mars conditions, defined in this paper to be 210 K and 3.7×10^{-2} Pa (=10 precipitable micrometers (pr μm) H₂O), the regolith materials held about 2.1 kg m⁻³ of H₂O. This calculation assumes that the specific surface area of the Martian regolith materials is 17 m² g⁻¹ [Ballou *et al.*, 1978] and that the density of the regolith is 1300 kg m⁻³.

Peak diurnal fluxes from the regolith, in the basalt model, were of the order of 6.5×10^{-5} kg m⁻² s⁻¹, meaning that the upper centimeter of regolith had both the capacity and the dynamic capability to fully supply the 10 pr μm of H₂O to the atmosphere in a very short time. The flux is determined by the gradient between the surface (determined by adsorptive equilibrium in numerical models) and the base of the atmosphere (defined as the H₂O abundance at the lowest atmospheric grid point in numerical models), coupled with the wind velocity.

$$\Phi = \rho_a C_H U_* (\bar{r}_a - \bar{r}_s) \quad (2)$$

Where U_* is the shear velocity, C_H is the heat transfer coefficient, ρ_a and r_a are the air density and water vapor mixing ratio, and r_s is the water vapor mixing ratio at the planet's surface. The latter is the value determined by adsorptive equilibrium.

The occurrence of ground ice, both near the surface (on diurnal timescale) and at greater depth, was predicted to be suppressed by adsorption onto basaltic materials. It is always true that ground ice formation is retarded by adsorption, and the efficacy of this process is determined primarily by the adsorptive capacity of the regolith materials. Materials with high-adsorptive capacity segregate more H₂O molecule into lower energy adsorption sites, thereby suppressing the vapor pressure and slowing the approach to the frost point in a cooling regolith. The basalt isotherms previously used were very efficient at retarding ice formation. Numerical models of the diurnal boundary layer run with a basalt regolith never predicted ground ice at night, except near the poles in the winter hemisphere.

2.2. Comparison With Recent Measurements

In the course of measuring the simultaneous adsorption of H₂O and CO₂ on Mars, Zent and Quinn [1995] made the first measurements of adsorption of H₂O onto geologic materials at conditions appropriate to Mars. We measured the data at conditions that were considerably different than those used in previous measurements (Table 1) and which overlapped the range of partial pressures actually experienced by particulates on the Martian surface. We found that water did not displace CO₂ from adsorption sites very effectively.

Because the specific surface area of our palagonite sample was 94.01 m² g⁻¹, considerably higher than the area of Fanale's [1971] basalt sample, direct comparison of the data is not as

illuminating as comparison on a surface density basis. Further, because the adsorptive coverage is a sensitive function of temperature and pressure, and because we gathered data at conditions different than Fanale and Cannon's [1971, 1974], direct comparison of the data is even more questionable.

To facilitate a direct comparison, both basalt and palagonite data sets were separately fit to a simple adsorption isotherm which we have used in the past:

$$\rho_a = A_s \delta P^\gamma T^\beta \quad (3)$$

The term A_s represents the Brunauer, Emmett, and Teller (BET) surface area of the materials. It is included in this isotherm under the assumption that the adsorption is linear with surface area. Evidence that this is a valid assumption is discussed below. By fitting the new palagonite data and the old basalt data to the same form of (3) we permitted direct comparison of the data sets. The technique for finding δ , γ , and β is described by Zent *et al.* [1987].

We calculated the adsorptive coverage, on a per meter basis, for both the basalt and palagonite, under standard Mars conditions. The results of this calculation were surprising. We found that H₂O covered palagonite at a density of 4.88×10^{-9} kg m⁻² and that it covered basalt at a density of 1.87×10^{-7} m⁻². Under identical conditions the difference in the predicted density of coverage was a factor of 38. We considered this an unacceptable result.

In previous cases, where adsorption data for CO₂ were compared on a variety of adsorbents [Zent *et al.*, 1987], it was found that the mineralogy of the adsorbent was a second-order control. The density of CO₂ adsorbed onto basalt and palagonite differed by no more than a factor of 3; we were hard pressed to imagine why adsorbate density would differ by a factor of 30 for H₂O. As a simple check, we used (1), the original isotherm presented for basalt, and found approximately the same result, implying that the fit to (3) is not the cause of the discrepancy.

One clue to the origin of the problem is that a comparison of H₂O adsorbent densities calculated for basalt and palagonite, under the assumption that the conditions were 260 K and 50 Pa, produced surface density estimates that differed by less than a factor of 4, a result consistent with our expectations from the same experiment using CO₂. At 260 K and 50 Pa, both isotherms are being used to interpolate between measured adsorption data. By contrast, at $T = 210$ K and $P = 3.72 \times 10^{-2}$ Pa, the basalt isotherm is being used to extrapolate more than 2 orders of magnitude in pressure and 40 K in temperature; the palagonite isotherm is still being used in an interpolation regime. The distinction is due to the fact that the palagonite data were acquired at Mars-like conditions. It was our goal to resolve this uncertainty with respect to the role of mineralogy and surface area in the adsorption of H₂O on Mars.

2.3. Adsorption on Heterogeneous Surfaces

A planetary regolith such as Mars' is composed of a variety of minerals and poorly-crystalline solids. Pore sizes vary according to the mineralogy and sedimentary history of each grain. All realistic models of the adsorptive behavior of the Martian regolith must take this heterogeneity into account.

The surface of a solid of complex chemical composition contains many exposed atoms and functional groups, which interact in various ways with adsorbent molecules. Surface imperfections of molecular dimensions such as dislocations,

vacancies, flaws, cracks, and chemical contaminants also may be a major source of heterogeneity [e.g., Jaroniec and Madey, 1988]. The presence of strongly bound impurities on the adsorbent surface may also influence its adsorption properties significantly. These factors also play a role in influence in the heterogeneity of microporous solids; however, the major source of heterogeneity in microporous solids is their complex porous structure, which contains micropores of different dimensions and forms.

During adsorption on an adsorbent possessing heterogeneity the energetic state of an adsorbate molecule depends on its contact point with the surface. These adsorption sites correspond to minima of the adsorption potential. The adsorbed molecule is considered to be localized on the surface when it is held at the bottom of a potential well with a depth that is much greater than the thermal energy of the molecule (kT) (Figure 1).

Fundamental descriptions of adsorption onto solid surfaces rely upon the notion of the continuous energy distribution of the adsorption sites, $F(\epsilon)$, which is the derivative of the number of adsorption sites with respect to the adsorption energy. This function is characteristic of each adsorbate-adsorbent pair and cannot be predicted in advance but must be evaluated from adsorption data. It describes the range of adsorption energies in the adsorbate-adsorbent interaction. In real systems the function is continuous and usually has at least one peak, and often more. Nonetheless, a discrete distribution instead of a continuous distribution simplifies the evaluation of $F(\epsilon)$ from the experimental adsorption isotherm and is often assumed.

One can derive an analytical form of the adsorption isotherm $\theta(P, T)$ for a given form of $F(\epsilon)$ assumed a priori, or one can derive $F(\epsilon)$ for an adsorption isotherm assumed a priori. One can also derive numerical values of $F(\epsilon)$ for an experimental isotherm. The first theoretical formulation of the simplest case of the localized adsorption model was made by Langmuir [1918], who assumed that adsorbed molecules formed a localized monolayer, that the solid surface of the adsorbent is energetically homogeneous, and that lateral attractive interaction in the monolayer could be neglected.

The behavior of real adsorption systems does not obey the isotherms developed for models that assume energetic homogeneity of the adsorbent surface. More successful equations correspond to distributions that more accurately represent the majority of adsorption systems.

2.4. History of Extrapolation to Mars Conditions

The assertion that the adsorbate abundance increases with surface area has long been understood [Puri and Murari, 1963].

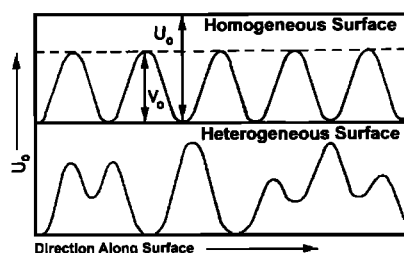


Figure 1. Cartoon of the variations in potential minima along (A) a homogeneous surface and (B) a heterogeneous surface. The quantity V_0 is the energy required for surface diffusion. After Jaroniec and Madey [1988].

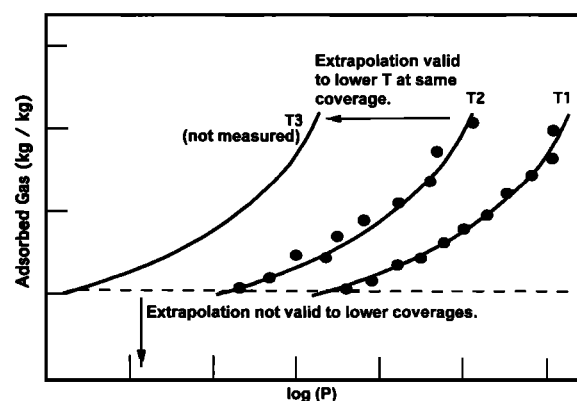


Figure 2. The limitation of the Anderson *et al.* [1967] technique for extrapolating adsorption isotherms. As long as data are available at some coverage, it is possible to estimate the P-T coordinates at which the same coverage will occur. It is not possible to extrapolate to coverages different than those covered in the data because the energy distribution function of the surface cannot be predicted.

Anderson and Tice [1973] found that normalizing unfrozen water content by specific surface area unifies the results from different soil samples. They found no more than a factor of 5 difference in adsorbed density (measured as kg H₂O per m² soil surface area) compared to a factor of several hundred on per mass basis (kg H₂O kg⁻¹ soil).

Anderson *et al.* [1967] made a numerical estimation of the amount of adsorbed water on Mars. They began with adsorption data from montmorillonite [Mooney *et al.*, 1952] and calculated the partial molar enthalpy of adsorption as a function of adsorptive coverage ($\Delta H(\alpha)$). Then, assuming that $\Delta H(\alpha)$ did not vary significantly with T , they calculated via the Clausius Clapeyron equation, the partial pressure at lower temperatures than but identical coverages to their data. Anderson *et al.*, [1978] tested the validity of this extrapolation technique. They took Mooney *et al.* [1952] data on sodium montmorillonite at 0° and 20°C and calculated $\Delta H(\alpha)$ from the Clausius-Clapeyron equation. Then they measured adsorption onto sodium at -5°C and compared their results to predictions based on $\Delta H(\alpha)$. They found good agreement. The critical aspect of their calculation is that it can be applied to extrapolate to lower temperatures, based on the generally valid assumption that $\Delta H(\alpha)$ is a weak function of temperature. However, it cannot be used to extrapolate to lower coverages than measured because $\Delta H(\alpha)$ cannot be predicted (Figure 2). Extrapolation of adsorption isotherms to lower coverages than measured depends upon an accurate understanding of $F(\epsilon)$. In practice, $\Delta H(\alpha)$ is the macroscopic thermodynamic quantity that relates to the energy frequency distribution of adsorption sites $F(\epsilon)$.

For all previous numerical modeling of H₂O adsorption on Mars, authors have assumed an empirical form of the adsorption isotherm to permit extrapolation to Martian conditions. This is a necessity because the data are too sparse to determine the adsorption isotherm unambiguously, and the available data have heretofore been at pressures and temperatures not fully appropriate to Mars. However, the isotherms commonly used in the Mars community do not account for any heterogeneity of the adsorbent surface. It was our hypothesis that the discrepancy between the measured palagonite adsorbate density

and the predicted basalt adsorbate density would be resolved such that the basalt would also have a low adsorbate density, and that the error in previous predictions derived from the use of a homogeneous adsorption isotherm in extrapolating to coverages lower than available in the laboratory.

3. Laboratory Data

We chose to test our hypothesis by measuring the adsorption of water onto basalt and montmorillonite at the same Mars-like conditions used for palagonite. We did not have access to the same samples on which the earlier basalt and montmorillonite adsorption data were gathered, however, those samples are not necessary. Our hypothesis was that the measured adsorbate densities would be within a factor of 5, regardless of mineralogy, as opposed to the factor of 38.

We collected samples of Page Mill basalt, a small Miocene eruption on the San Francisco Peninsula. The Page Mill basalt has been subjected to a small amount of hydrothermal alteration in its lower half, where pyrite and marcasite are found as alteration products. Our samples came from the upper half of the remaining flow and do not have visible secondary minerals.

Samples were cut on a diamond saw, and an internal sample was placed in a ball mill and ground under N₂. The sample was transferred to a glove bag and sieved under N₂ as well. We used the <34 μm fraction, sealing the sample before removing it from the glove bag.

Our montmorillonite sample was a source clay from the Clay Mineral Society (CMS) (SWy-1), a Na-montmorillonite from the Newcastle Formation in Crook County, Wyoming. The CMS source clays are homogeneous and collected in batches of over 500 kg, assuring long-term access to a standard clay. SWy-1 has been extensively studied [e.g., *Koster van Groos and Guggenheim*, 1984; *Hall and Astill*, 1989]. It was used as received.

For surface area measurements of the basalt an aliquot of the sample was pumped overnight while heating to 150°C to produce the cleanest surface possible. The sample was then cooled to 78 K and adsorption measurements were made using N₂. The pressure drop technique was used with pressure measurements made with a capacitance manometer. The resulting N₂ isotherm can be used to derive the specific surface area via the BET technique [*Brunauer et al.*, 1938]. We found that the N₂ specific surface area was $1.801 \times 10^4 \text{ m}^2 \text{ kg}^{-1}$, as compared to $9.8 \times 10^3 \text{ m}^2 \text{ kg}^{-1}$, which was the surface area of the *Fanale and Cannon* [1971] sample.

A similar approach could not be used on SWy-1 because the interlayer sites in this smectite clay are available only to the polar H₂O molecule and not to N₂. The surface area of the clay was determined by the glycerol method, which can measure interlayer sites to be $6.63 \times 10^5 \text{ m}^2 \text{ kg}^{-1}$ [*van Olphen and Fripiat*, 1979]. By contrast the N₂ specific surface area of the same sample, found via the BET technique, is $3.18 \times 10^4 \text{ m}^2 \text{ kg}^{-1}$. The differences in these values underscores the fact that specific surface area is an operationally defined quantity, which cannot always be directly interpreted in geometric terms. A given technique measures only the density of sites compatible with the given probe molecule.

The adsorption measurements then proceeded on a separate aliquot of the sample. It was pumped overnight at 120°C to remove all exchangeable gases. Although this preconditioning will not remove surface hydroxyls which may affect adsorptive behavior, the fresh surfaces of our basalt sample should be

altered little. For montmorillonite the sample was pumped for 24 hours because of kinetic barriers involved in equilibrating interlayer sites with their environment.

The samples were cooled to 209 K, and a pressure drop experiment was performed with He in order to quantify the dead volume in the sample. The sample was then pumped for 1 hour to remove all He. Thereafter the sample was exposed to H₂O vapor buffered at 204.9 K, equivalent to a partial pressure of 0.34 Pa. Equilibration was allowed to proceed for 4 hours for the basalt and 24 hours for SWy-1. The samples were then isolated from the H₂O source. To quantify the adsorbate load, the samples were heated to 120°C to drive off the water and pumped through an LN₂ trap to collect all the H₂O. After 4 hours of trapping the LN₂ trap is isolated from the system and heated to 200°C to drive all H₂O into the vapor phase. An injection is then made from the LN₂ trap into the He carrier gas stream, leading to a gas chromatograph. The gas stream was fed into a chromatographic column and through a thermal conductivity detector. The area under the thermal conductivity curve is integrated to determine the total abundance of H₂O in the stream. The column temperature is maintained at 140°C to prevent condensation or adsorption of water.

Our measurements indicated that the abundance of adsorbed water on our Page Mill basalt sample was $2.9 \times 10^{-4} \text{ kg kg}^{-1}$, on a sample with $1.801 \times 10^4 \text{ m}^2 \text{ kg}^{-1}$; the resulting density of adsorbed molecules is $1.6 \times 10^{-8} \text{ kg m}^{-2}$. For SWy-1 we measured $1.85 \times 10^{-2} \text{ kg H}_2\text{O kg}^{-1}$ on a sample with specific surface area of $6.63 \times 10^5 \text{ m}^2 \text{ kg}^{-1}$; the resulting density of adsorbate is $2.8 \times 10^{-8} \text{ kg m}^{-2}$. The abundance of H₂O on palagonite at the same conditions is $1.1 \times 10^{-8} \text{ kg m}^{-2}$; hence the mineralogic variations of the adsorbent accounts for a factor of about 2.5. By contrast the previously used isotherm, (1), corrected for the difference in surface area, predicts for these conditions that the abundance of adsorbed H₂O would be $7.9 \times 10^{-3} \text{ kg kg}^{-1}$, or a density of $4.8 \times 10^{-7} \text{ kg m}^{-2}$. The discrepancy between the predicted and measured abundances is a factor of almost 30 (cf. the factor of 38 difference between the predicted basalt adsorbate density and the measured palagonite density). Therefore we believe that the actual adsorbate density at Mars-like conditions is 15 to 30 times lower than previously believed and varies by no more than a factor of a few regardless of mineralogy.

4. Extrapolation to Mars

4.1. Search for New Isotherms

Having found that the apparent discrepancy in the adsorptive capabilities of basalt and palagonite derive from extrapolation of an unrealistic isotherm, it is now required that a more suitable isotherm be produced. Fortunately, because the chemical engineering industry is dependent on adsorption for many of processes, there are an almost limitless number of models for adsorption which account for the heterogeneity of the adsorbent surface. The task is to find a fit between the available data and the available models of adsorption.

We selected three adsorption models to test against the data; the Freundlich isotherm

$$\theta = \left(\frac{K^*P}{1 + K^*P} \right)^n \quad (4)$$

the Langmuir-Freundlich isotherm

$$\theta = (K^*P)^n / [1 + K^*P]^n \quad (5)$$

Table 2. Best Fits of Palagonite and Montmorillonite Data to Isotherms

Mineral	Type	K_0	ε	ν	χ^2
Palagonite	Freundlich	7.54×10^{-9}	2697.2	0.4734	19.35
Montmorillonite	Toth	2.207×10^{-9}	4157.1	0.6593	16.51

and the Toth isotherm

$$\theta = K^*P/[1 + (K^*P)^\nu]^{1/\nu} \quad (6)$$

In all of these isotherms,

$$K^* = K_0 \exp(\varepsilon/T) \quad (7)$$

where K_0 is a measure of the fundamental frequency of the molecular vibrations, and ε is a measure of the energy of the adsorptive bond. The term ν is the heterogeneity parameter, which describes the shape of $F(\varepsilon)$. The process of fitting the data to these isotherms consists of finding K_0 , ε , and ν , such that some quantity which measures goodness of fit is minimized. By returning the goodness of fit parameter we will have a basis for comparing the best fit for each type of isotherm, and we will select the best single isotherm for each data set individually.

We selected these three isotherms because they correspond to a range of shapes that the energy frequency distribution might assume. For example, (5) corresponds to $F(\varepsilon)$ skewed strongly toward lower energies, whereas (6) represents a symmetrical $F(\varepsilon)$, and (7) represents an overabundance of high-energy adsorption sites. There is no a priori way to know whether other isotherms might better represent the data, except to check each isotherm individually. Since we do not have an actual sample of Mars, but terrestrial analogs, we concluded that it would be sufficient to survey a variety of isotherms without an exhaustive attempt to examine all isotherm models.

We chose to implement the fit of the data to these multidimensional and nonlinear isotherms, which contain in practice a term for the specific surface area of the adsorbents as well, via the Levenberg-Marquardt approach [Press *et al.*, 1988].

We have N data points to be fit to a nonlinear model which has M adjustable variables. The data can be represented by

$$\alpha = \alpha(P, T, A_s; K, \varepsilon, \nu) \quad (8)$$

where P is pressure, T is temperature, A_s is the specific surface area of the soil, and these three variables control the amount of adsorbed H₂O. The parameters K , ε , ν are adjusted until the χ^2 parameter

$$\chi^2 = \sum_{i=1}^N \left(\frac{\alpha_i - \alpha(P, T, A_s, K, \varepsilon, \nu)}{\sigma_i} \right)^2 \quad (9)$$

is minimized. Unless otherwise reported, we assumed that the standard deviation associated with each data set was 50%. Although a 50% standard deviation is in some cases higher than evinced in the data, we chose to use this value because (1) the palagonite data, acquired through a successive series of trappings, seems to have such a high standard deviation [Zent and Quinn, 1995], (2) hysteresis is evident in all adsorption isotherms, so the actual adsorbed amount can seldom be predicted with much better precision, regardless of the accuracy of the data, and (3) the partial pressure and adsorbed load vary over a few orders of magnitude so that a 50% uncertainty in

the adsorbed load is not devastating to our predictive ability as it might seem.

The technique requires the partial derivative of the isotherm to be fit, with respect to each of the variables K , ε , and ν . The user provides initial values of the variables to be fit, and the technique walks downhill until it finds a local minimum, and reports the corresponding values of K , ε , and ν . The technique also produces an estimate of χ^2 so that goodness of fit can be compared quantitatively. It is often necessary to restart the procedure with different initial values in order to be sure it is not getting stuck in a local minimum.

There is no way from a purely mathematical point of view to be certain that there is not a better fit in some corner of parametric space than the best fit returned by the routine. However, adsorption theory provides additional constraints on the parametric space; ν must be between 0 and 1. Moreover, ε is effectively a measure of the molar energy of adsorption, so its value is constrained as well.

The data that we examined were (1) the palagonite data reported by Zent and Quinn [1995], (2) the montmorillonite data reported by Mooney *et al.* [1952] and Anderson *et al.* [1978], along with our measurement, and (3) the basalt data of Fanale and Cannon [1974] and our measurement reported above. Because no single data set covers the full range of Mars conditions, except for palagonites, we were compelled to combine data from different samples of the same adsorbent. In the case of montmorillonite the clays are all Na-montmorillonites from Wyoming and were specifically selected to be standard clays, and further, specifically selected to be similar to their predecessors [van Olphen and Fripiat, 1979]. One might argue that the data from three similar but distinct montmorillonite clays might be more representative of putative Martian smectites than a single sample.

In the case of basalt we were unable to fit the available data to any of our isotherms. All of the adsorption isotherms that we tested above are derived for adsorptive coverages less than or equal to one monolayer. In cases where we have data at Mars-like conditions the coverage is always less than one monolayer. However, the basalt database gathered by Fanale and Cannon [1974], at conditions warmer and wetter than standard Mars conditions, yielded coverages in excess of one monolayer for all but two data points. The molecules which are adsorbing onto the basalt are sampling not the silicate surface but the modulation of the electronic signature of that surface by the first monolayers of H₂O. Combined with our single measurement, only three data points are available, at coverages less than one monolayer, to fit the equations above. Not enough data exist at the appropriate conditions to make a numerical estimate of the actual adsorption isotherms of terrestrial basalt under Mars-like conditions.

The fits returned by the Levenberg-Marquardt technique, along with the corresponding values of χ^2 for palagonite and montmorillonite, are shown in Table 2. It can be seen that the best fit for the palagonite data is the Freundlich isotherm.

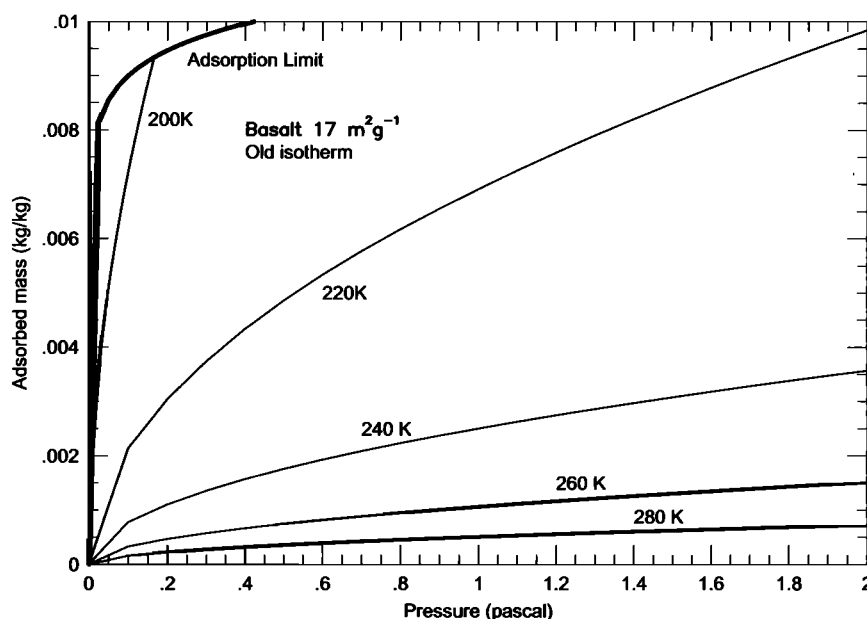


Figure 3. The adsorption of H₂O predicted by (1). This is the isotherm that has been used in the literature for many years.

Indeed, the χ^2 value is considered to indicate an excellent fit if it is of the order of $N-M$. In the case of the palagonite, $N = 23$ data points and $M = 3$ adjustable parameters. Value $\chi^2 = 19.35$, indicating an excellent fit.

We have chosen to use the adsorption data of *Mooney et al.* [1952] and *Anderson et al.* [1978] in our fitting exercise, in spite of the fact that the data were not gathered at Mars-like conditions. Our rationale for this choice is that virtually all of the data represent coverage of less than a single monolayer, based on surface area measurements. Therefore the H₂O molecules are serving as surface probes of the silicate surface. The adsorption isotherms against which we are testing the data are equipped to describe this type of adsorption. For montmorillonite the best fit is from the Toth isotherm, with $\chi^2 = 16.51$ for $N = 25$ and $M = 3$.

4.2. Adsorption

The simplest and most direct comparison of the adsorption differences between basalt and palagonite have already been made. For a given specific surface area at standard Mars conditions (10 pr μm H₂O, 210 K) the density of H₂O adsorbate on a basalt surface, its clay weathering products, or the weathering products of its glass, is approximately 30 times lower than previously estimated, based on (1). We can expand this somewhat by considering the effects of variable surface area.

The actual surface area of the Martian regolith materials are very poorly constrained. The only estimates, based on in situ measurements, were made by *Ballou et al.* [1978], who looked at the gases released from Martian surface materials in the Viking Gas Exchange experiment. On the basis of the release of atmospheric gases, with the exception of O₂, they estimated that the specific surface area of one sample at the VL1 site was $1.7 \times 10^4 \text{ m}^2 \text{ kg}^{-1}$. Figures 3 and 4 compare the new palagonite and old basalt adsorption isotherms under the assumption that the specific surface area of the sample is $1.7 \times 10^4 \text{ m}^2 \text{ kg}^{-1}$. If the regolith materials are assumed to have a density of 1300 kg m^{-3} , then 1 m^3 of basalt, using (1), would have held

2.99 kg of exchangeable H₂O adsorbate. On the other hand, a cubic meter of palagonite, using (5), and assuming the constants in Table 2, would hold only 0.083 kg of exchangeable H₂O.

An independent consideration of the montmorillonite mineralogy and surface area is not as trivial. Clay weathering products are always very fine grained, particularly in the case of smectite clays, in which interlayer sites are available to water molecules because of their high degree of polarity. We show the montmorillonite adsorption isotherm in Figure 5, under the assumption that the specific surface area is $1 \times 10^6 \text{ m}^2 \text{ kg}^{-1}$, which is the value reported for the Wyoming bentonite by *Mooney et al.* [1952]. The surface area is the most important variable in these estimates. A small amount of smectites will dominate the adsorptive behavior of the regolith because it will dominate the surface area.

4.3. Diffusion

One effect of adsorption on the role of the regolith in modulating the Martian climate is that it serves to slow the process of vapor diffusion. This is important because the ability of the regolith to buffer the atmospheric H₂O column is mediated by the rate at which H₂O can diffuse through the regolith. The physical explanation of this effect is that diffusing molecules spend considerably more time adsorbed onto the walls of the pores than in actual diffusion.

In order to account for the slowing of diffusion, the effective diffusion coefficient through the background gas, in this case CO₂, must be scaled by the slope of the diffusion coefficient with respect to the number density of H₂O molecules in the gas phase (γ). We can see the magnitude of this effect on the diffusion coefficient by comparing the scaling factor ($d\alpha/d\gamma$) for basalt and palagonite, as in Figure 6. It is clear that the greatest difference is at the lowest temperatures, which is consistent with the fact that the basalt adsorption data of *Fanale and Cannon* [1971] were correct for the temperature range in

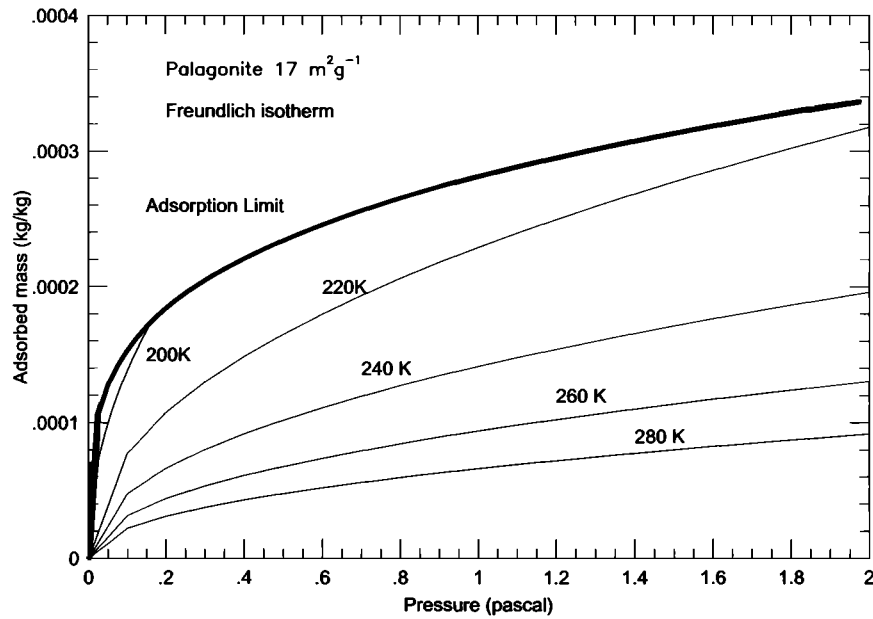


Figure 4. The adsorption of H₂O predicted by (5) and which we fit to adsorption data on palagonite in this study. Notice the differences in the coordinates of the y axis.

which they were measured. It is only the extrapolation to the lowest temperatures that led to error.

Therefore the distinction between the isotherm presented here and those used previously is keenest at low temperature. In Table 3 we have calculated the adsorption at appropriate temperatures and where the effective length scale for diffusion is $\delta = (2tD_f)^{1/2}$ and $t = 8.874 \times 10^4$ s, Mars' rotation period. It is clear that the difference in diffusion scale length is significant only at night in the near-surface regolith. In addition, the mathematical expression for $d\alpha/d\gamma$ contains the sur-

face area A_s as a coefficient, so the scaling factor will vary linearly with area.

4.4. Boundary Layer

With the stipulation that the specific surface area remain fixed, it is clear that the buffering capacity of the regolith must be reevaluated in order to account for the lower density of adsorbate discussed here. We have amended a model of H₂O exchange in the Martian boundary layer [Zent *et al.*, 1993] to account for the use of the palagonite adsorption isotherm (5)

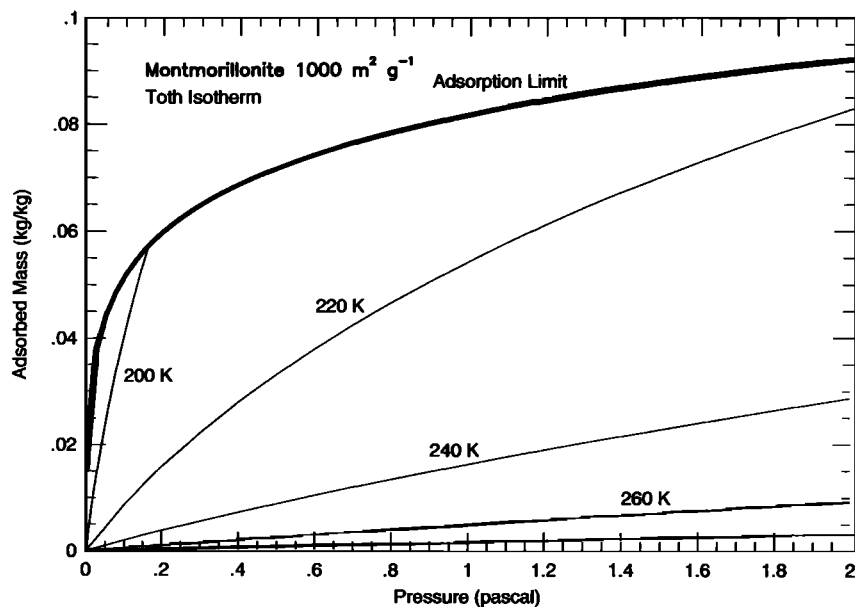


Figure 5. The adsorption of H₂O onto montmorillonite predicted by (7) from our fit. Because the surface area of the montmorillonite is so high, the adsorbate load is also higher. On a surface density basis it differs from palagonite and basalt by less than a factor of 2.5.

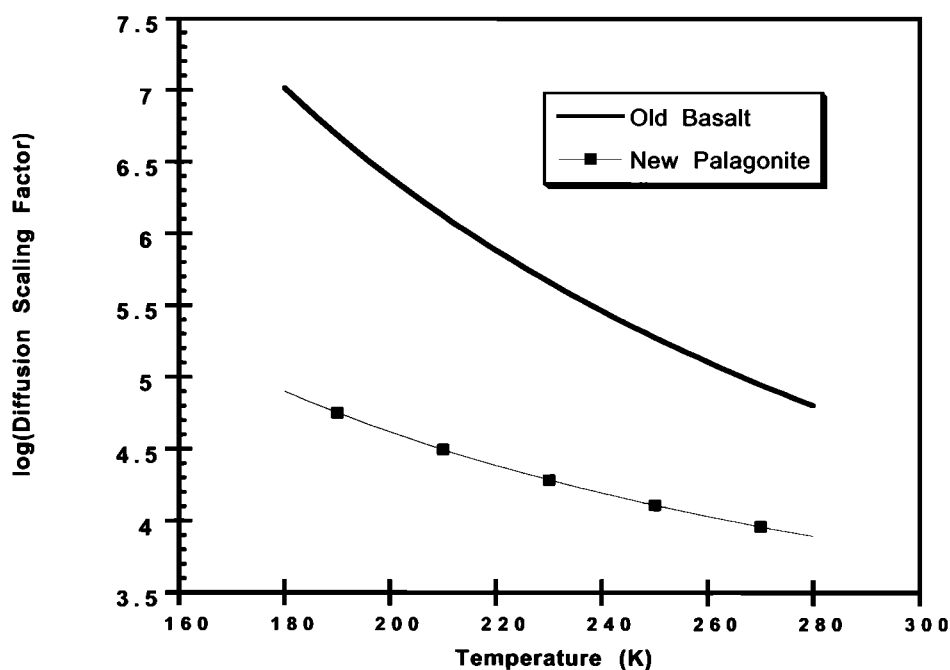


Figure 6. The differences in scaling of the diffusion coefficient between (1) and (5). The diffusion coefficient must be divided by this number so that the effective diffusion coefficient is now thought to be larger than before.

and the montmorillonite adsorption isotherm (7). Using the same baseline case that was used in our earlier paper for the basalt isotherm, we find that the exchange of H₂O between the surface and atmosphere is considerably muted for the non-smectites and enormously enhanced for montmorillonite. For example, it was mentioned earlier that the peak fluxes in the diurnal exchange between the atmosphere and basalt regolith were of the order of $6.5 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$, which occurred near local noon when the regolith was heated and desorbing H₂O rapidly. The new model of the adsorptive exchange between the atmosphere and regolith now suggests that the peak fluxes are of the order of perhaps $2 \times 10^{-6} \text{ kg m}^{-2} \text{ s}^{-1}$, which means that the entire atmospheric column of H₂O, (roughly 0.01 kg m^{-2} for $10 \text{ pr } \mu\text{m}$), will require almost an hour and a half to exchange with the regolith. By contrast a similar calculation using the old basalt isotherm suggested that the entire atmospheric column of H₂O could exchange with the diurnal skin depth of the regolith in only 3 min. The peak flux for montmorillonite is of the order of $3 \times 10^{-4} \text{ kg m}^{-2} \text{ s}^{-1}$. The predicted flux histories for each of these cases is explored in Figure 7. It is worth noting that we assume adsorptive equilibrium is constantly maintained; this might not be a valid assumption for montmorillonite because the kinetics of exchange with interlayer sites may be too slow. A separate investigation of the kinetics of H₂O adsorption on smectites under Martian conditions is being carried out.

An even further barrier to rapid buffering is the reduced capacity of the regolith. The effective damping depth for diffusion through the regolith is

$$\delta = \sqrt{2Dt} \quad (10)$$

where D is the effective diffusion coefficient. The effective diffusion coefficient, when scaled for adsorption, is the bulk diffusion coefficient (around $10^{-4} \text{ m}^2 \text{ s}^{-1}$ for H₂O through

CO₂ under Mars-like conditions), divided by the factor shown in Figure 6. Assuming $T = 250 \text{ K}$, representative of the temperature in the uppermost centimeter of regolith during the warmest part of the day, the scaling factor would be about $10^{5.2}$. Therefore, for the old basalt isotherm, the effective diffusion coefficient D would be around $5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$, and the corresponding effective damping depth, that is, the thickness of regolith that would be able to exchange H₂O with the atmosphere diurnally is about 1 cm. Within the upper centimeter of basaltic regolith the total mass of H₂O is roughly $0.023 \text{ kg of H}_2\text{O m}^{-2}$, more than twice the total atmospheric column.

For the palagonite calculations presented here the adsorptive slowing of diffusion is somewhat less, and the effective scale length of diffusion at 250 K is about 3.3 cm. Therefore more of the regolith can exchange its adsorbed H₂O load with the atmosphere diurnally than in the basalt case. However, the upper 3.3 cm of regolith holds only $2.5 \times 10^{-3} \text{ kg m}^{-2}$, or about 25% of the atmospheric column. The adsorbed H₂O contours are shown in Figure 8 for each of these cases.

Indeed, the population of adsorbed molecules available to the atmosphere diurnally will always increase with surface area, even though the effective diffusion coefficient will always decrease, as long as the diffusion is not in the Knudsen regime.

Table 3. Comparison of Diffusion Coefficients and Length Scales

Hour	New Isotherm		Old Isotherm	
	$D_f, \text{ m}^2 \text{ s}^{-1}$	$\delta, \text{ m}$	$D_f, \text{ m}^2 \text{ s}^{-1}$	$\delta, \text{ m}$
1200	1.4×10^{-9}	1.5×10^{-2}	3.8×10^{-9}	2.6×10^{-2}
2400	6×10^{-11}	3.3×10^{-3}	9×10^{-13}	4×10^{-4}

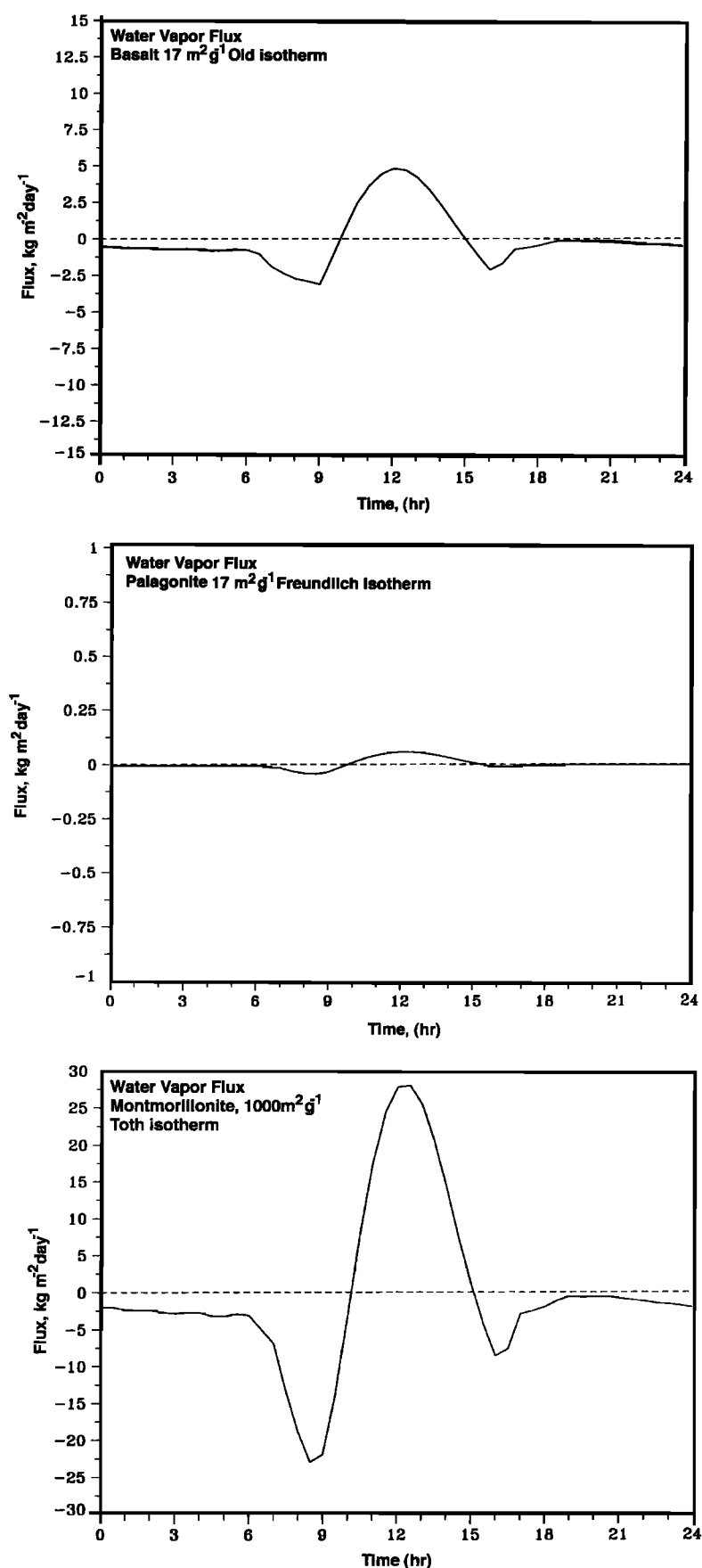


Figure 7. The diurnal water vapor flux from (a) the previously used isotherm (1), (b) the palagonite isotherm, and (c) the montmorillonite isotherm.

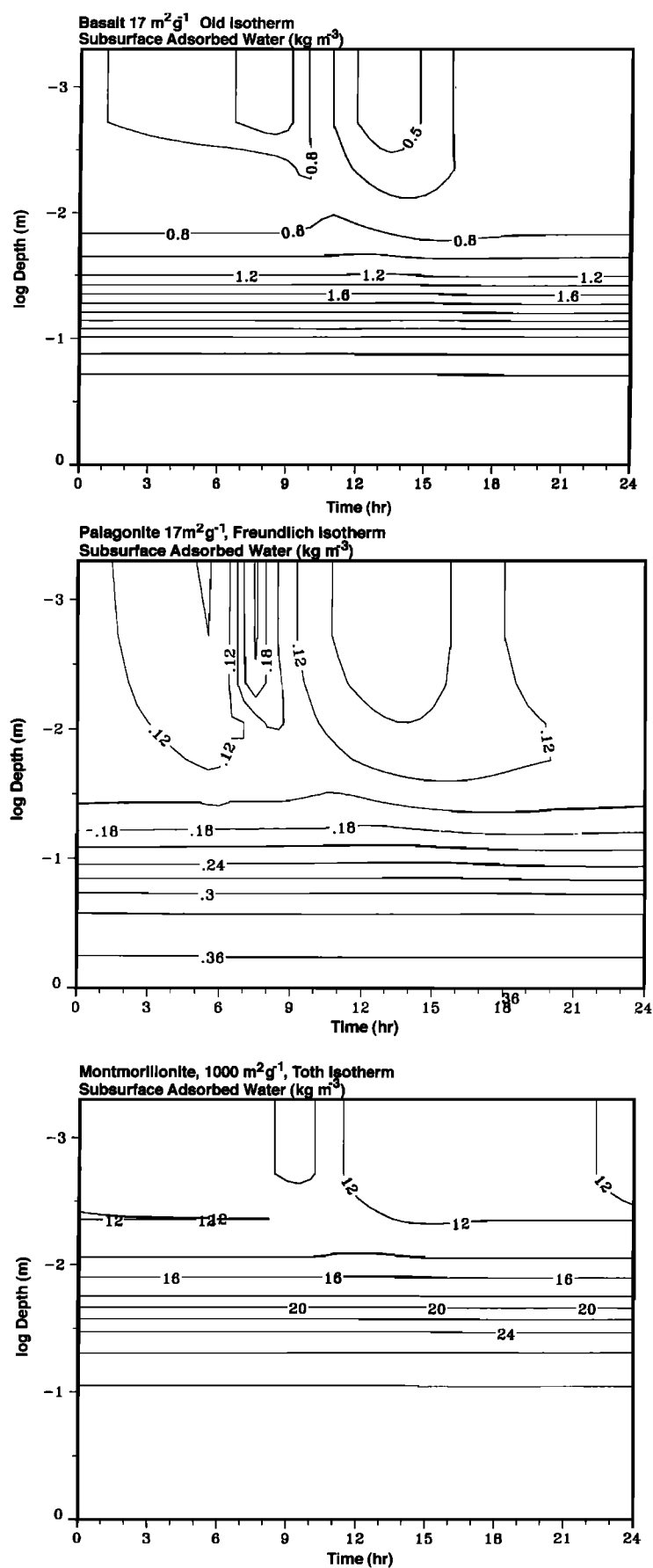


Figure 8. Contours (a) 96.0, (b) 96.1, and (c) 96.2 of diurnal variation in subsurface adsorbed H₂O. Units are kg m⁻³.

Therefore the ability of the local regolith to buffer the atmosphere on a diurnal basis will depend upon the amount of clay-sized materials present. This conclusion is distinct from previous understanding of the role of the Martian regolith, which held that any material would have sufficient capacity to buffer the diurnal atmosphere. Houben *et al.* [this issue] argue that it is possible, using the results of this work, to make a very crude estimate of the effective particle size of the uppermost few centimeters of the Martian regolith. By assuming the adsorptive behavior described in this paper and tuning the results of their global circulation model to match Viking observations, they concluded that the effective specific surface area is of the order of $7.5 \times 10^4 \text{ m}^2 \text{ kg}^{-1}$. If the effective surface area is much higher, water convected off the polar cap in the spring is removed from the atmosphere before reaching lower latitudes. If the area is much lower, the atmosphere saturates and ice deposits form over much of the planet. Using the specific surface areas of $2 \times 10^4 \text{ m}^2 \text{ kg}$ for nonsmectites and $1 \times 10^6 \text{ m}^2 \text{ kg}$ for smectites, we could estimate that the globally averaged abundance of smectites in the Martian regolith would be of the order of 5%, with the remaining 95% of the regolith materials being nonsmectites. These proportions are not ruled out on spectroscopic grounds and are consistent with adsorption models and the models of the annual Martian water cycle. Given the uncertainties in the foregoing calculation, there is obviously little quantitative significance to such an estimate, aside from the fact that a moderately high specific surface area is required and that at least a small percentage of smectite is probably required.

5. Summary

New measurements of the adsorption of H₂O onto terrestrial minerals, under Mars-like conditions, disagree with predictions made on the basis of the most frequently used adsorption isotherms. We report here on additional measurements, which confirm that previous estimates of H₂O adsorptive coverage of Martian surface minerals were overestimated. We have analyzed this discrepancy as a result of extrapolating an empirical expression for adsorption that can be shown to contain unrealistic assumptions. We show that new isotherms, developed to describe adsorption on heterogeneous surfaces, are well suited to describing adsorption at Mars-like conditions. The new isotherms predict that the adsorptive capacity of the regolith is lower than previously thought, if normalized for available surface area, that vapor diffusion is more rapid, and that ground ice must occur more commonly than previously thought.

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